Class 12

- For general two player zero sum game, finding the max-min strategies and the value of the game can be solved as linear programming problem.
  See Solan's book, Section 5.2.5

Theorem For a two player zero sum game,

(σ_1^*, σ_2^*) is a max-min strategy profile $\Rightarrow$ (σ_1^*, σ_2^*) is an NE

Proof Suppose v is the value of the game

$$U(\sigma_1^*, \sigma_2^*) \geq v, \quad U(\sigma_1^*, \sigma_2^*) \leq v$$

$$\Rightarrow \quad U(\sigma_1^*, \sigma_2^*) = v$$

We need to show $U(\sigma_1^*, \sigma_2^*) \geq U(\sigma_1, \sigma_2^*) \quad \forall \sigma_1 \in \Sigma_1$ --- (1)

($U(\sigma_1^*, \sigma_2^*) = v$) and

$U(\sigma_1^*, \sigma_2^*) \leq U(\sigma_1^*, \sigma_2) \quad \forall \sigma_2 \in \Sigma_2$ --- (2)

($U(\sigma_1^*, \sigma_2^*) = v$)

$U(\sigma_1, \sigma_2^*) \leq v$ σ_2^* is a max-min strategy P-2

This gives us (1).

For (2),

$U(\sigma_1^*, \sigma_2) \geq v$ σ_1^* is a max-min strategy for P-1

So, (σ_1^*, σ_2^*) is a NE.

Theorem Suppose (σ_1^*, σ_2^*) is an NE of a two-player zero sum game. Then value exists in mixed strategies and (σ_1^*, σ_2^*) is a max-min strategy profile.

Proof

$$\underbrace{U(\sigma_1^*, \sigma_2^*)}_{=v} \geq U(\sigma_1, \sigma_2^*) \quad \forall \sigma_1 \in \Sigma_1 \quad \text{---(i)}$$

$$\underbrace{U(\sigma_1^*, \sigma_2^*)}_{=v} \leq U(\sigma_1^*, \sigma_2) \quad \forall \sigma_2 \in \Sigma_2 \quad \text{---(ii)}$$

From (ii), $\underline{U(\sigma_1^*, \sigma_2) \geq v} \quad \forall \sigma_2 \in \Sigma_2$

$\Rightarrow \underline{v} \geq v$

From (i) $U(\sigma_1, \sigma_2^*) \leq v \quad \forall \sigma_1 \in \Sigma_2$

$\Rightarrow \bar{v} \leq v$

$\bar{v} = \min_{\sigma_2 \in \Sigma_2} \max_{\sigma_1 \in \Sigma_1} U(\sigma_1, \sigma_2)$

$\bar{v} \geq \underline{v}$

$\Rightarrow v = \underline{v} = \bar{v}$

The value exists.

From (i) and (ii), we have (σ_1^*, σ_2^*) is a max-min strategy profile.

- For a game G and its mixed extension Γ,

$$\underline{v}_i = \max_{\sigma_i \in \Sigma_i} \min_{\sigma_{-i} \in \Sigma_{-i}} U_i(\sigma_i, \sigma_{-i})$$

Theorem

Suppose σ^* is an NE

$$U_i(\sigma^*) \geq \underline{v}_i \quad \forall i \in N$$

Proof: Assume for contradiction $U_i(\sigma^*) < \underline{v}_i$.

Suppose σ_i is a max-min strategy for player i

$$U_i(\sigma_i, \sigma_{-i}) \geq \underline{v}_i \quad \forall \sigma_{-i} \in \Sigma_{-i}$$

$\Rightarrow U_i(\sigma_i, \sigma_{-i}^*) > U_i(\sigma^*)$, a contradiction to the fact that σ^* is an NE.

Computation of NE

		2: F (y)	2: D ($1-y$)
1: F (x)		(2,1)	(0,0)
1: D ($1-x$)		(0,0)	(1,2)

$$br_i(\sigma_{-i}) = \{\bar{\sigma}_i \mid U_i(\bar{\sigma}_i, \sigma_{-i}) \geq U_i(\sigma_i, \sigma_{-i}) \ \forall \sigma_i \in \Sigma_i\}$$

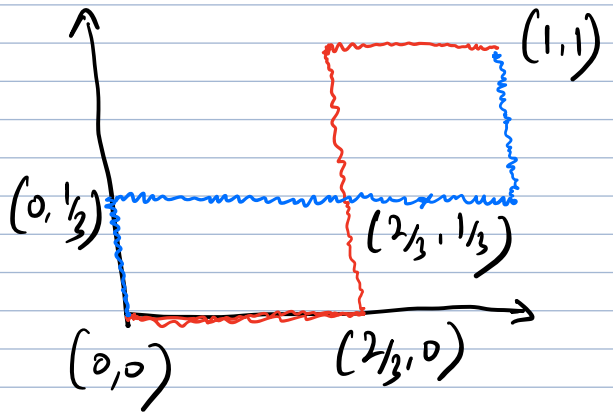
A strategy profile is σ^* an NE if $\sigma_i^* \in br_i(\sigma_{-i}^*)$ $\forall i$

$$U_1(\sigma_1, \sigma_2) = U_1(x, y) = 2xy + 1(1-x)(1-y)$$
$$= 2xy + 1 - x - y + xy$$
$$= 3xy + 1 - x - y$$
$$= x(3y-1) + (1-y)$$

$$br_1(y) = \begin{cases} 0 & y < \frac{1}{3} \\ [0,1] & y = \frac{1}{3} \\ 1 & y > \frac{1}{3} \end{cases}$$

$$U_2(x, y) = 1xy + 2(1-x)(1-y)$$
$$= 3xy + 2 - 2x - 2y$$
$$= y(3x-2) + (2-2x)$$

$$br_2(u) = \begin{cases} 0 & u < \frac{2}{3} \\ [0,1] & u = 2/3 \\ 1 & u > 2/3 \end{cases}$$



The intersecting points of these two graphs are the NE of the game.

- Theorem: Suppose σ^* is an NE. Suppose for player i

$$\sigma_i^*(s_i^1) > 0, \quad \sigma_i^*(s_i^2) > 0$$

for some $s_i^1 \in S_i$ and $s_i^2 \in S_i$

$$U_i(s_i^1, \sigma_{-i}^*) = U_i(s_i^2, \sigma_{-i}^*)$$

- Corollary: For a Nash Equilibrium σ^*,

$$U(\sigma_i^*, \sigma_{-i}^*) = U_i(s_i, \sigma_{-i}^*)$$

$\forall s_i \in S_i$ s.t. $\sigma_i^*(s_i) > 0$